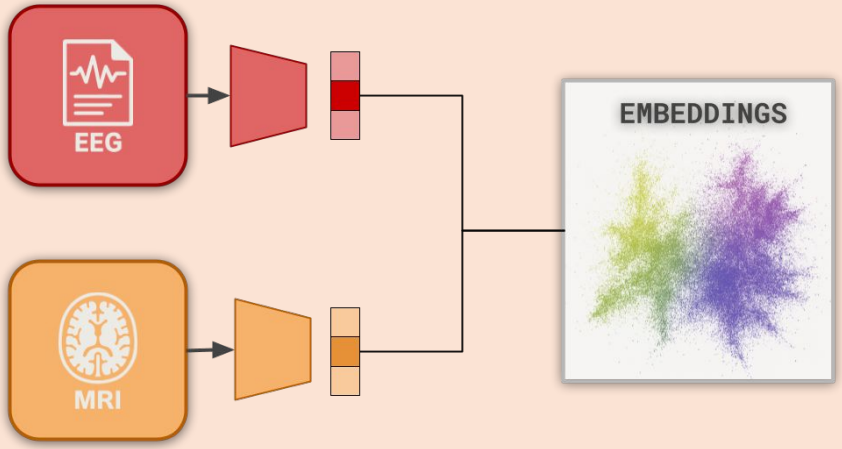
SigLIP in Federated Learning

Bryan Cardenas
Douwe van der Wal
High Performance Machine Learning

SURF

SigLIP Training of Embeddings

$$\mathcal{L} = -\frac{1}{|B|}\sum_{i=1}^{|B|}\sum_{j=1}^{|B|}\log\frac{1}{1+e^{z_{ij}(-tx_i\cdot y_j+b)}}$$



Based on CLIP with Sigmoid Loss

Modality Specific Encoder

Create a shared embedding space

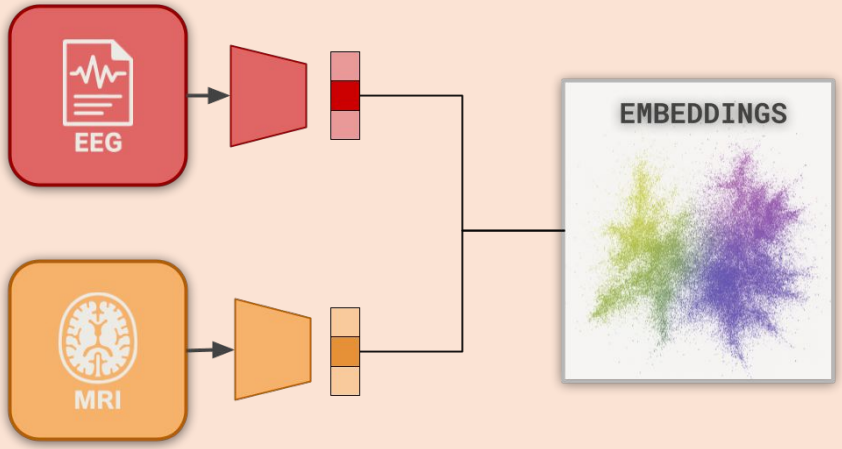
Distributing SigLIP model relatively straightforward

[2303.15343] Sigmoid Loss for Language Image Pre-Training

SURF

SigLIP Training of Embeddings

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Based on CLIP with Sigmoid Loss

Modality Specific Encoder

Create a **shared embedding space**

Distributing SigLIP model relatively straightforward

[2303.15343] Sigmoid Loss for Language Image Pre-Training

SURF

How to Federate?

01.

Not all clients have the same patient data

Data distribution varies a lot

02.

Not all clients have the same modalities

Vertical modalities

03.

Not the same amount of data

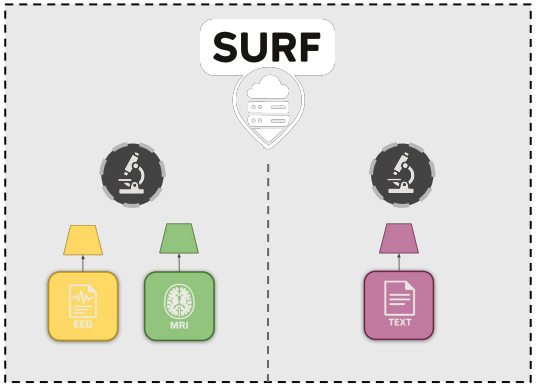
One representation type would dominate over the others

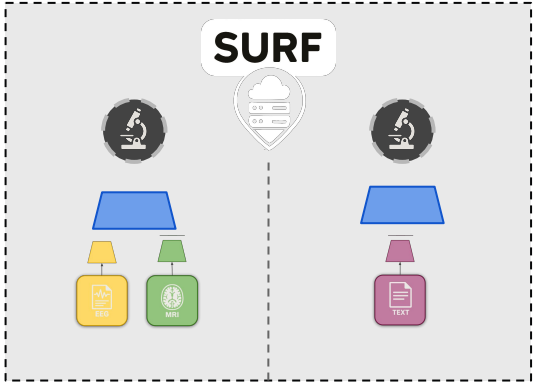
04.

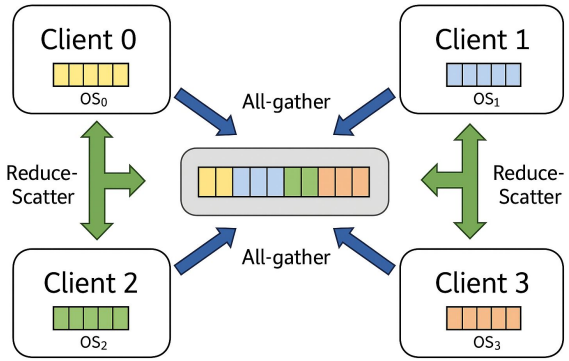
How to partition the model?

How to *share* or break up the
model for the benefit of all?

Federated SigLIP: Assume no shared (patient) data points







Distribute the Encoders

Each client gets a different "specialized" encoder

Share the embeddings or the loss?

Unified Encoder and One Shared Modality

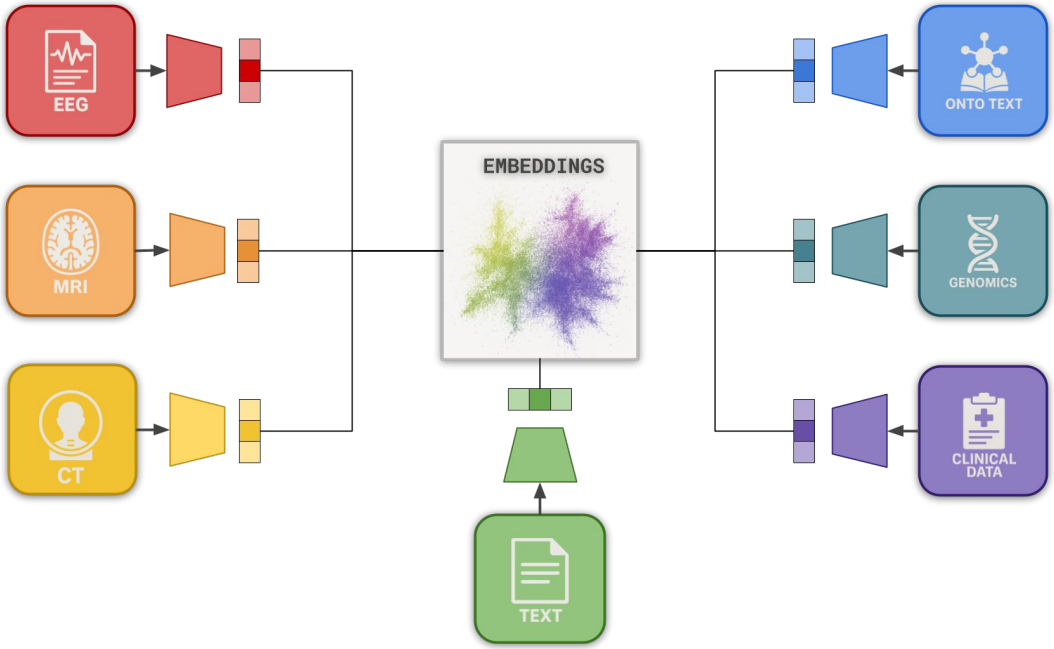
Every client should have one modality present at all times, for instance, text.

Communicating parts of the Network

Communicate gradients? Losses? Embeddings? Or merge them in the global client? How many updates?

SigLIP for True Multi Modality

$$\mathcal{L} = -\frac{1}{|B|}\sum_{i=1}^{|B|}\sum_{j=1}^{|B|}\log\frac{1}{1+e^{z_{ij}(-tx_i\cdot y_j+b)}}$$



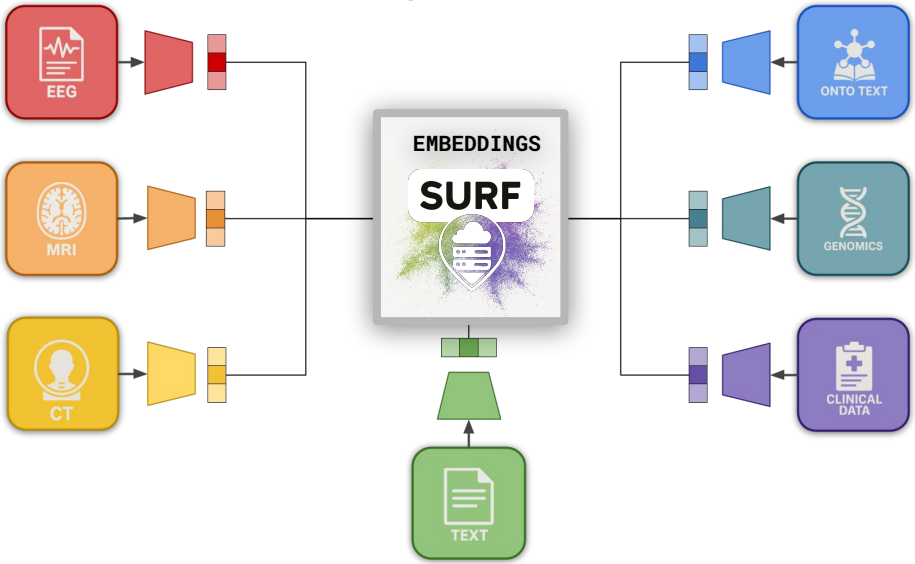
Best if there is one “binder” modality

Modality Specific Encoders

Best with bigger batch sizes

SigLIP for True Multi Modality

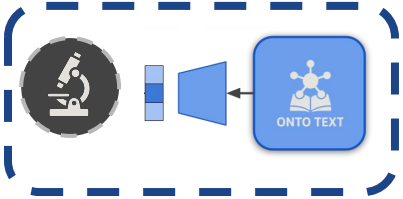
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Algorithm 2 Federated SigLIP-style Training

1: Train encoders on public data using SigLIP at server

SigLIP for True Multi Modality

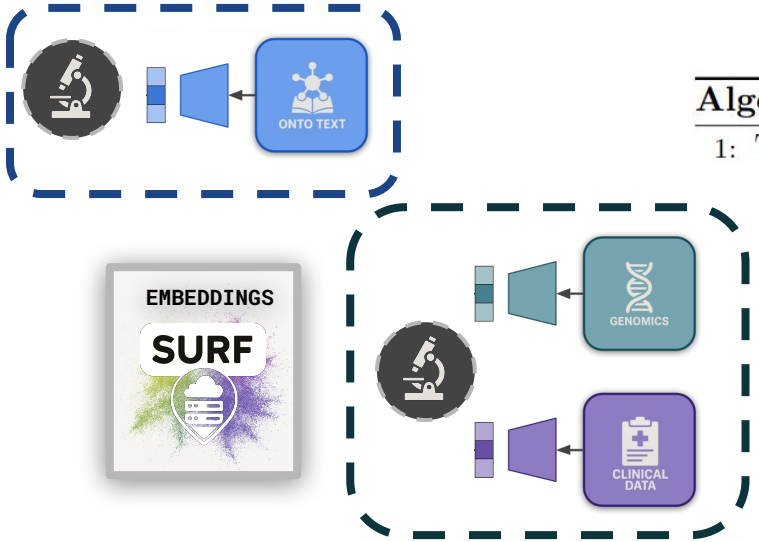




Algorithm 2 Federated SigLIP-style Training

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for round $r = 1$ to N **do**
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    for all clients k **do**
        Train locally on private data

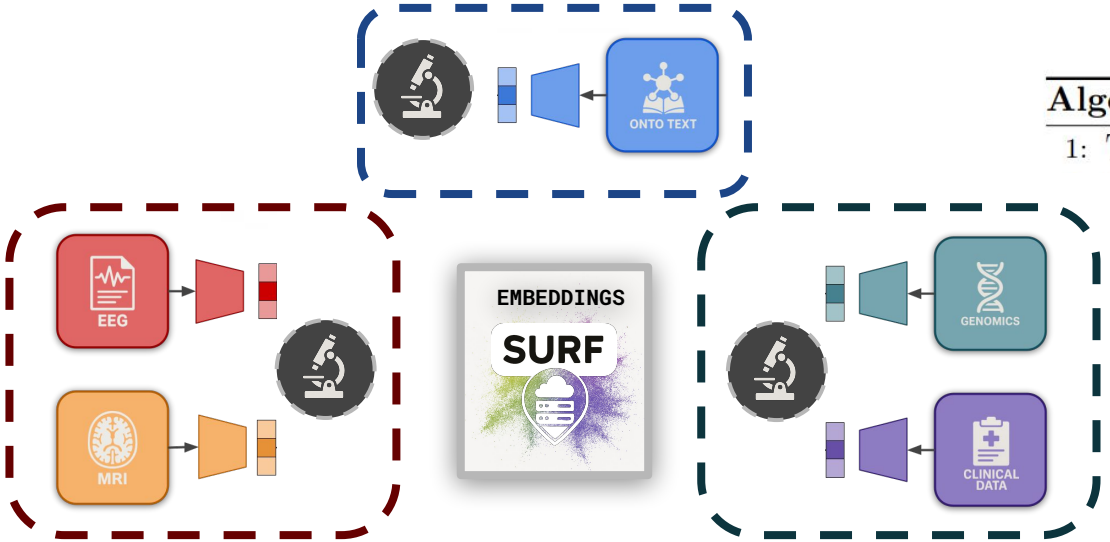
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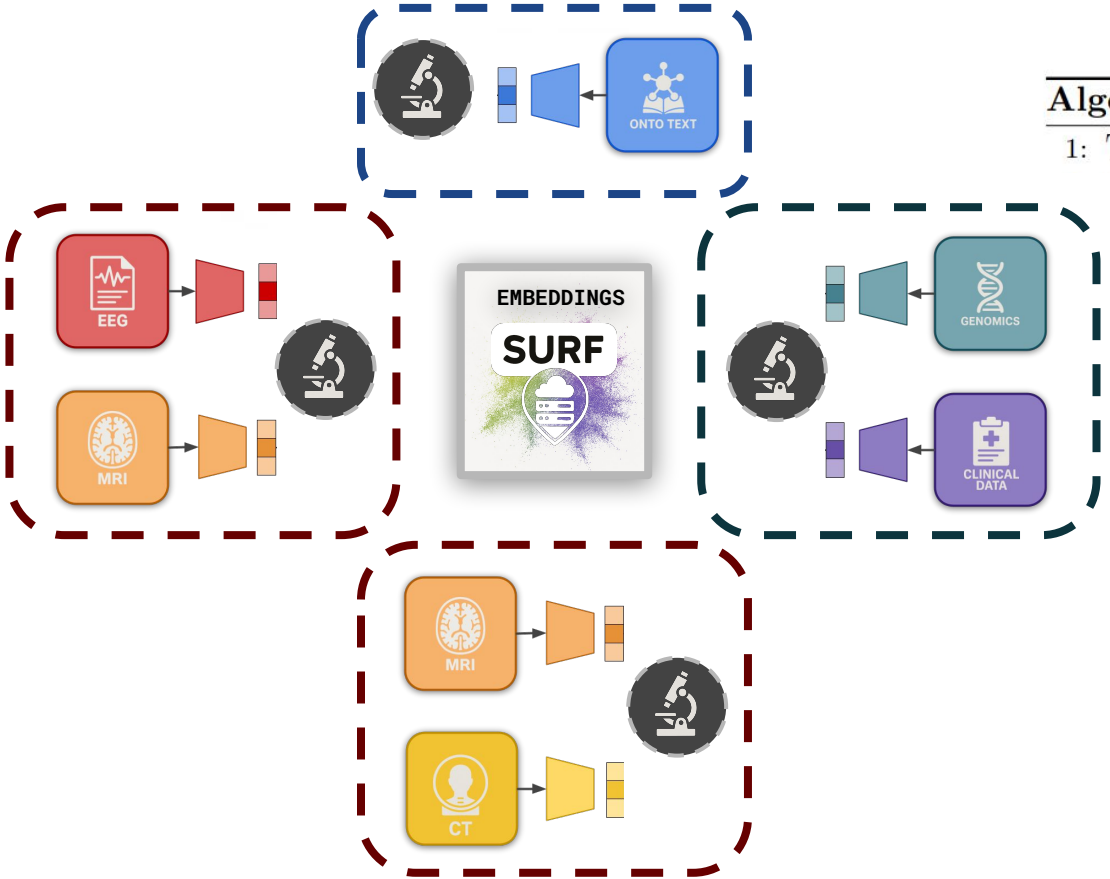
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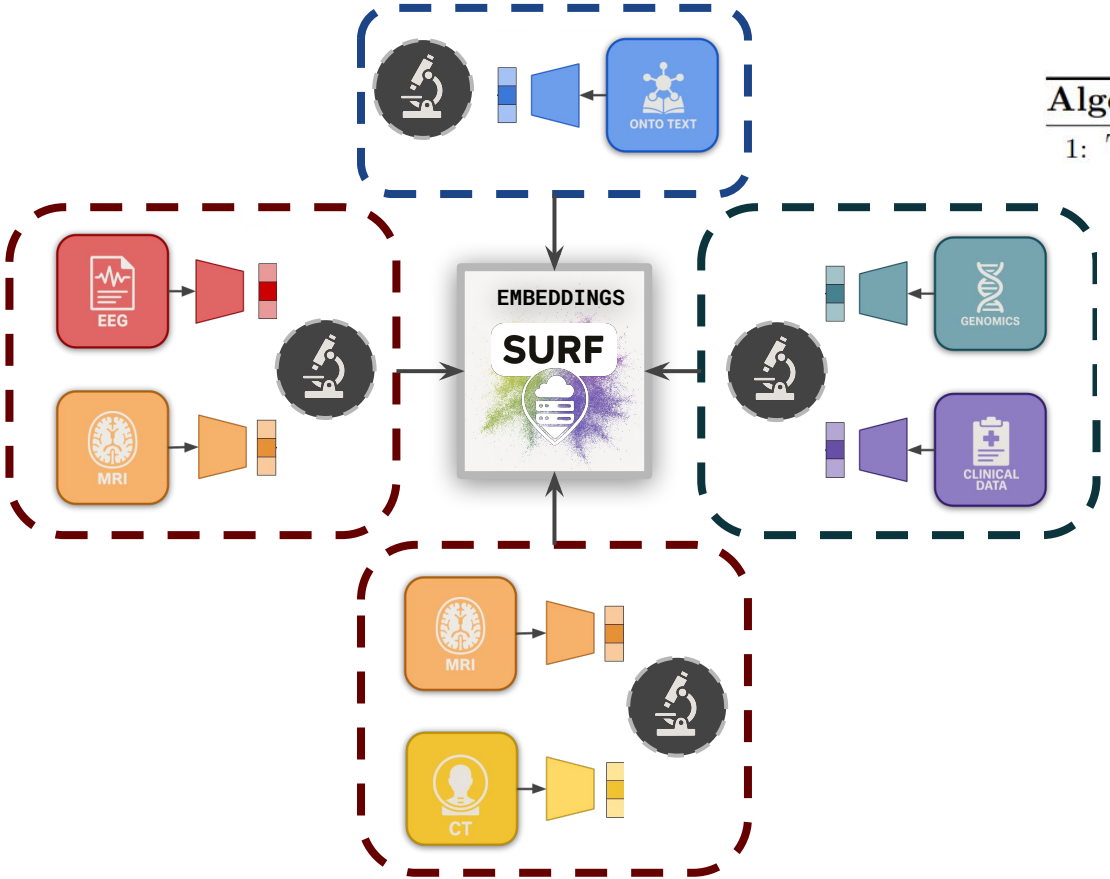
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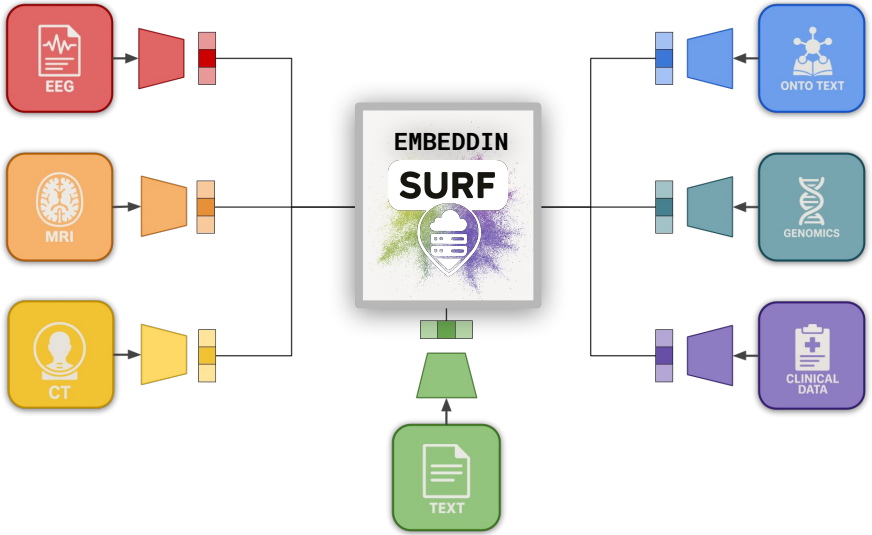


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5: Train locally on private data
6: Send updates to server and aggregate

SigLIP for True Multi Modality

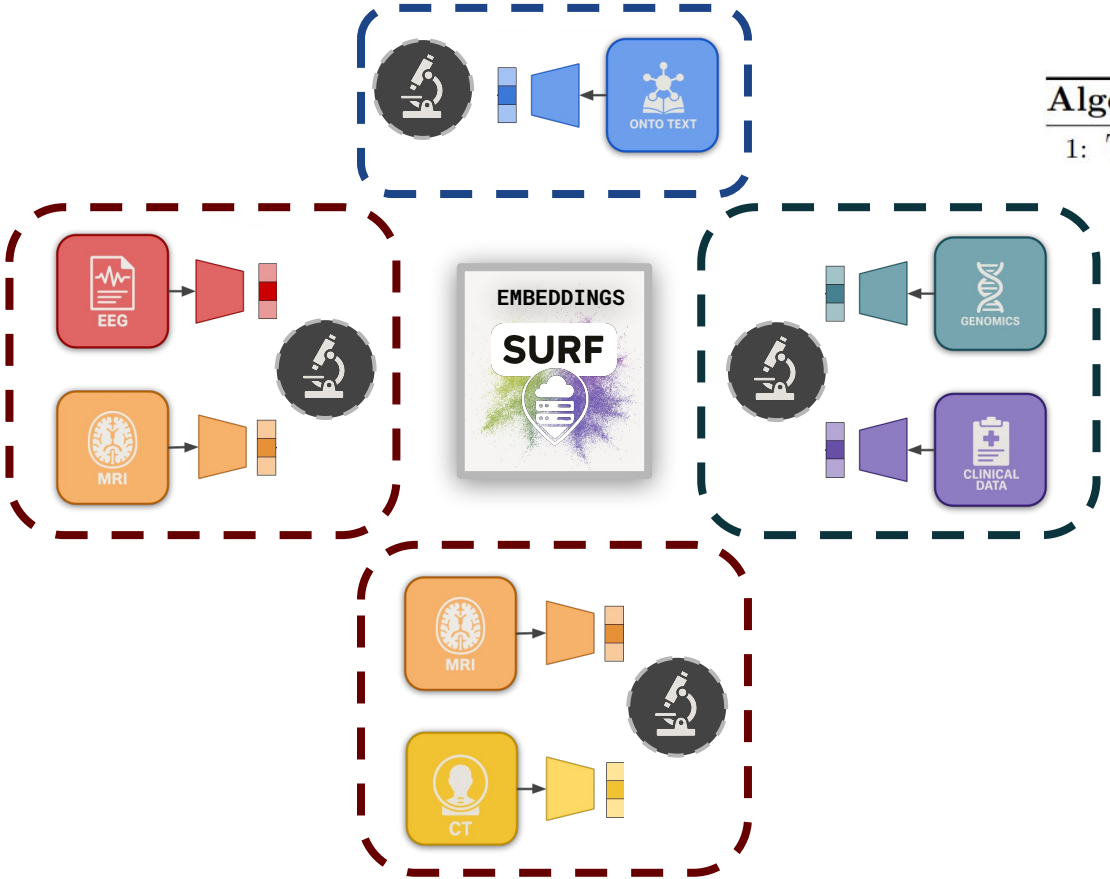
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        Send updates to server and aggregate
    end for
    Aggregate updates at server

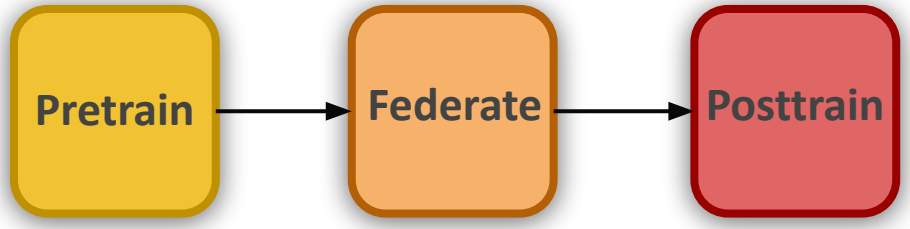
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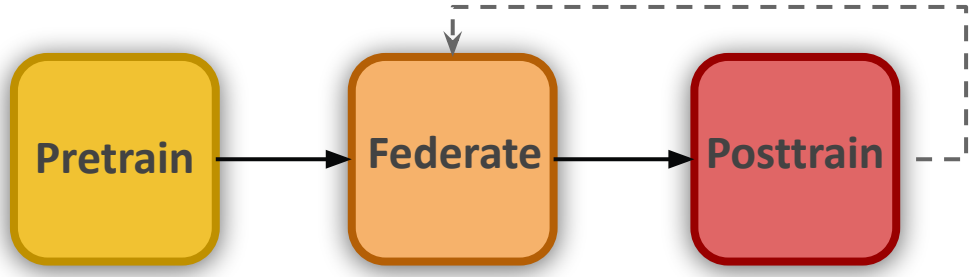
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end for
(Optional) Fine-tune centrally

SigLIP for True Multi Modality



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Suggestions?